**4.**

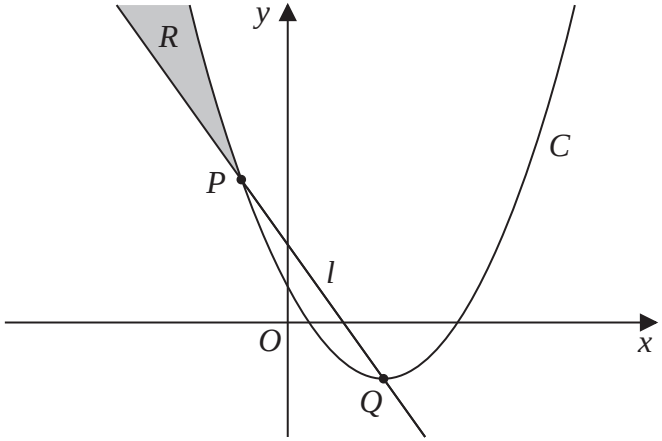



**Figure 2**

The points $P$ and $Q$, as shown in Figure 2, have coordinates $(-2, 13)$ and $(4, -5)$ respectively.

The straight line $l$ passes through $P$ and $Q$.

(a) Find an equation for $l$, writing your answer in the form $y = mx + c$, where $m$ and $c$ are integers to be found.

**(3)**

The quadratic curve $C$ passes through $P$ and has a minimum point at $Q$.

(b) Find an equation for $C$.

**(3)**

The region $R$, shown shaded in Figure 2, lies in the second quadrant and is bounded by $C$ and $l$ only.

(c) Use inequalities to define region $R$.

**(2)**